Alvin Banh

669-216-7629 | albanh@mit.edu | linkedin.com/in/alvin-banh | github.com/alvin-banh

Education

Massachusetts Institute of Technology — Cambridge, MA
Bachelor of Science in Computer Science — *June 2027*

Relevant Coursework *Machine Learning, Software Construction, Design and Analysis of Algorithms, Fundamentals of Programming, C and Assembly, Robotics*

Experience

Undergraduate Researcher — June 2024 - February 2025
MIT CSAIL — *Cambridge, MA*

- Collaborated with PhD students and faculty to evaluate **1200+ research papers** on foundation models
- Documented emerging trends in generative AI, large language models, and transformer architectures
- Presented findings weekly to guide future research initiatives and experimental priorities

Engineering Intern — July 2022 - August 2022
Bizlink Holding Inc — *Fremont, CA*

- Researched hardware architecture of **20+ cable and autonomous vehicle (AV) companies**
- Benchmarked **30+ key performance metrics** including processing power and latency for AI deployment
- Synthesized data into **10 comprehensive reports** of hardware and AI specifications for cross-functional teams

Projects

Deep Learning Model for RNA folding — December 2024 – Present

- Developing a high-performance AI model to determine secondary and tertiary structures of RNA
- Experimenting with transformer-based architectures inspired by AlphaFold-3 and BPfold for improved accuracy
- Implementing recently published AI models such as hierarchical reasoning models to progress computational biology research

Real-Time Translation Mask (Finetuned Multimodal LLM) — February 2024 – Present

- Engineered a wearable COVID-19 face mask performing live speech-to-speech translation
- Accelerated translation speed **3x** for streamlining live conversation
- Trained the AI model to handle multiple languages while **reducing training time by 20%**

COVID-19's Origins Logistic Regression Model — February 2021 – March 2021

- Analyzed **30,000+ genomic sequences** comparing COVID-19 and bat coronavirus strains to identify high-risk transmission regions
- Implemented feature reduction methods, achieving a **91% accuracy rate**
- Applied L1 regularization to fix overfitting, ensuring statistically significant predictions across 5-fold validation

Awards

Ruth and Norman Rales Scholar — March 2022
KIPP Public Charter Schools — *San Jose, CA*

- Awarded a competitive four-year national scholarship recognizing exceptional public service
- Demonstrated consistent dedication to academics by maintaining a 3.0 GPA or higher
- Strengthened leadership skills through active participation in annual conferences

Technical Skills

Programming Languages: Python, C++, Java, JavaScript, TypeScript, C, SQL
Frameworks & Libraries: React, Node.js, Express.js, Django, PyTorch, TensorFlow, scikit-learn
AI & ML: NLP, OpenCV, pandas, NumPy
Developer Tools: Git, VS Code, IntelliJ
Databases: MySQL, PostgreSQL, MongoDB